CTA Cardiac Excel to DICOM SR Configuration Instructions

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Prerequisites:

- Microsoft Excel desktop app
- DICOM SR mapping server (Hyland ModLink, Altamont Passport SR, etc.)
- Read/write access to network file share
- Local workstation connectivity to DICOM SR mapping server
- Modality Work List (MWL) for Calling AE Title

Setup:

Place the files on a network file share that is accessible to the technologists.

Example share:

```
\\hospital\files\WebForms
```

If the PACS supports External Links with variables linked to the selected exam, the workbook can be launched from PACS in context of the study. Apply quotes around linked variables that contain spaces (i.e., Patient Name, Study Description).

Example command:

```
C:\Windows\System32\cscript.exe \\hospital\files\WebForms\CTA\LaunchCTA.vbs
"%pn" %pi %an "%sd" %id
```

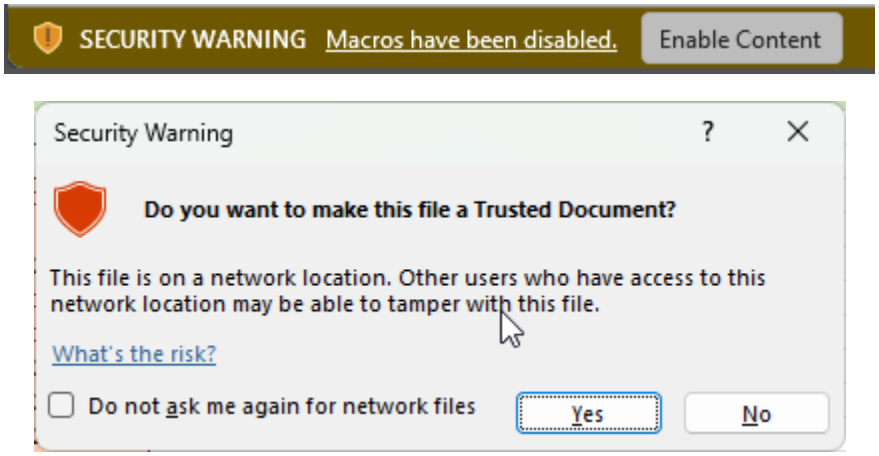
By default the LaunchCTA VB Script accepts 5 arguments, and each one is written to a specified cell (row, column) within the workbook upon opening.

```
Arg1 = Patient Name (Cell 1, 2)

Arg2 = MRN (3, 2)

Arg3 = Accession (5, 2)

Arg4 = Study Date (7, 2)

Arg5 = User ID (1, 9)
```

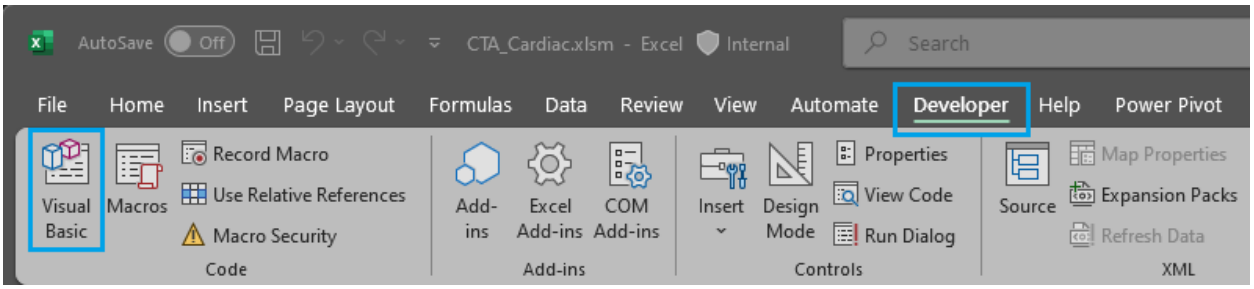
Modify **LaunchCTA.vbs** so that the **filePath** points to the **CTA_Cardiac.xlsm** workbook.

```
filePath = "\\hospital\files\WebForms\CTA\CTA_Cardiac.xlsm"
```

When the workbook is first opened on a workstation, there will likely be a Security Warning for disabled macros. Click “Enable Content” and then click “Yes” to make the file a Trusted Document. This needs to be set for each user on every workstation.

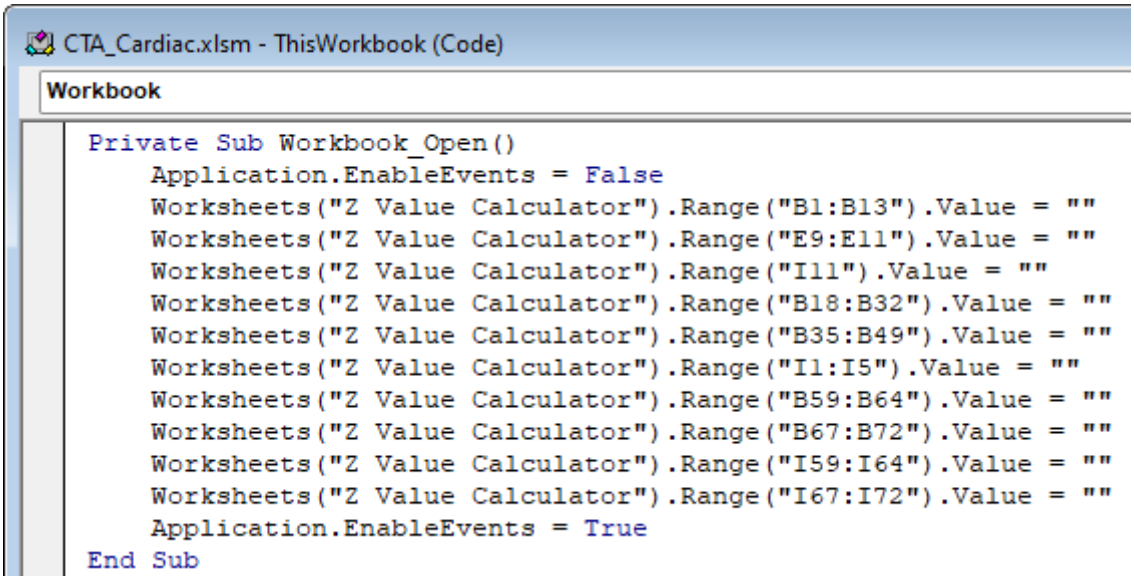


To access the workbook code, enable Developer in the ribbon (File → Options → Customize Ribbon) then click on **Visual Basic**.



Note the XML mapping can be viewed under **Source**.

The Visual Basic contains code for `Workbook_Open()` that temporarily disables all events while the cells are reset, effectively clearing the workbook from any data that may belong to another study.




```
Private Sub Workbook_Open()
    Application.EnableEvents = False
    Worksheets("Z Value Calculator").Range("B1:B13").Value = ""
    Worksheets("Z Value Calculator").Range("E9:E11").Value = ""
    Worksheets("Z Value Calculator").Range("I11").Value = ""
    Worksheets("Z Value Calculator").Range("B18:B32").Value = ""
    Worksheets("Z Value Calculator").Range("B35:B49").Value = ""
    Worksheets("Z Value Calculator").Range("I1:I5").Value = ""
    Worksheets("Z Value Calculator").Range("B59:B64").Value = ""
    Worksheets("Z Value Calculator").Range("B67:B72").Value = ""
    Worksheets("Z Value Calculator").Range("I59:I64").Value = ""
    Worksheets("Z Value Calculator").Range("I67:I72").Value = ""
    Application.EnableEvents = True
End Sub
```

After events are re-enabled, the arguments passed from the LaunchCTA script are written to their appropriate cells, prepopulating the workbook in context of the study and preventing the user from having to manually fill-in demographic and exam details.

Module1 contains the subroutine for the “Validate & Send” button. Modify the Modality Work List (**FINDSCU**) and SR mapping server (**STORESCU**) variables in the Configuration Block to the site environment. In most use cases the remaining code can be left as-is.

CTA_Cardiac.xlsm - Module1 (Code)

(General)

```
Sub ExportToModLink()
    ' ==========================================================================
    ' CONFIGURATION BLOCK - Update these settings for different environments
    ' ==========================================================================

    ' Network and DICOM Settings
    Const FINDSCU_IP As String = "172.30.98.114"        ' Target IP for C-FIND
    Const FINDSCU_PORT As String = "204"                ' Target Port for C-FIND
    Const FINDSCU_AET As String = "WEBFORM_CT"          ' Calling AE Title
    Const FINDSCU_AEC As String = "WORKGATE"            ' Called AE Title
    Const STORESCU_IP As String = "172.30.98.213"       ' Target IP for C-STORE
    Const STORESCU_PORT As String = "4096"              ' Target Port for C-STORE

    ' Excel Worksheet and Range Settings
    Const WS_MAPPINGS As String = "Mappings"            ' Sheet for validation
    Const WS_CALC As String = "Z Value Calculator"      ' Sheet containing Patient info
    Const RNG_NUM_CHECK As String = "J5:J9"             ' Range to check for numbers
    Const RNG_ACCESSION As String = "B5"                ' Cell with Accession Number
    Const RNG_PATIENT_ID As String = "B3"               ' Cell with MRN / Patient ID
    Const XML_MAP_NAME As String = "report_Map"         ' XML map object name

    ' ==========================================================================
    ' END CONFIGURATION BLOCK
    ' ==========================================================================
```

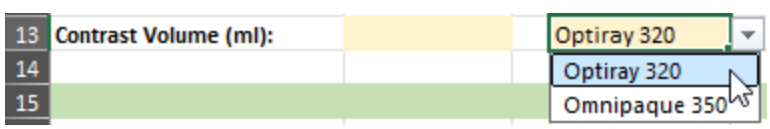
This script performs several operations:

1. Verify numeric values in the Num Table
2. Modality Work List C-FIND with DCMTK’s `findscu` to verify accession and MRN and ensure the procedure is reportable
3. Confirm data is exportable to XML and export as XML into the “Utils” network share subfolder
4. Convert XML to DICOM SR using DCMTK’s `xml2dsr` utility
5. Verify both XML and SR files were recently modified (within past minute) to confirm export and conversion success
6. Send DICOM SR to SR mapping server via DCMTK’s `storescu`
7. Save a copy of the workbook in network share subfolder “Submitted” with accession and timestamp as the file name

The **Mappings** worksheet contains all the information that will be written to the DICOM SR and auto populates based on the values from the Z Value Calculator worksheet. This is where the Custom Field names can be changed to match the desired nomenclature. If any cell placements are adjusted on Z Value Calculator, then its corresponding Mappings cell formula will need adjusted as well. It's recommended to leave the Mappings worksheet as-is and use the default field names, but if edits are needed, proceed with caution.

	A	B	C	D	E	F	G	H	I	J
1										
2		Report	Type	Comprehensive SR		Num Table				
3			SOP Class UID	1.2.840.10008.5.1.4.1.1.88.33						
4			SOP Value	ComprehensiveSRStorage		ns1:relationship	ns1:value	ns1:designator	ns1:meaning	ns1:value2
5		Device	Modality	SR		CONTAINS	PAT_HEIGHT	PAT1	PAT_HEIGHT	0
6			Manufacturer	CT Technologist		CONTAINS	PAT_WEIGHT	PAT2	PAT_WEIGHT	0
7			Model	Excel		CONTAINS	BSA1	BSA1	BSA1	0.00
8			Serial	2069872136		CONTAINS	HR	HR1	HEART_RATE	0
9			Version	1		CONTAINS	CONTRAST	CONT1	CONTRAST_VOL	0
10		Patient	ID	0		CONTAINS	BSA2	BSA2	BSA2	0.00
11		Study	Date	19000100						
12			Time							
13			Accession	0						
14			Description	CT CARDIAC V2						
15		Series	Number	999		Text Table				
16			Date	20260320						
17			Time	15:10		ns1:relationship	ns1:value	ns1:designator	ns1:meaning	ns1:value2
18			Description	CT Measurements SR		CONTAINS	IV_SITE	IV_SITE	IV_SITE	0
19		Instance	Number	1		CONTAINS	IV_SIZE	IV_SIZE	IV_SIZE	gauge
20		Document				CONTAINS	CONTRAST_MED	CONTRAST_MED	CONTRAST_MED	Optiray 320
21		Content				CONTAINS	RADIOLOGIST	RADIOLOGIST	RADIOLOGIST	n/a
22		Containe	Flag	SEPARATE		CONTAINS	SINUS_A_SYS	SINUS_A_SYS	SINUS_A_SYS	-
23		Concept	Value	125100		CONTAINS	SINUS_B_SYS	SINUS_B_SYS	SINUS_B_SYS	-
24			Designator	DCM		CONTAINS	SINUS_C_SYS	SINUS_C_SYS	SINUS_C_SYS	-
25			Meaning	CTA Cardiac Measurements		CONTAINS	STJ1_SYS	STJ1_SYS	STJ1_SYS	-
26		Containe	Flag	SEPARATE		CONTAINS	STJ2_SYS	STJ2_SYS	STJ2_SYS	-
27			Relationship	CONTAINS		CONTAINS	ASC1_SYS	ASC1_SYS	ASC1_SYS	-
28		Concept	Value	125100		CONTAINS	ASC2_SYS	ASC2_SYS	ASC2_SYS	-
29			Designator	DCM		CONTAINS	PTA1_SYS	PTA1_SYS	PTA1_SYS	-
30			Meaning	Aortic Pulmonary		CONTAINS	PTA2_SYS	PTA2_SYS	PTA2_SYS	-
31						CONTAINS	DTA1_SYS	DTA1_SYS	DTA1_SYS	-

The **Static** worksheet is only used to control the Contrast Type values seen in the dropdown of cell D13 in Z Value Calculator for easy selection.

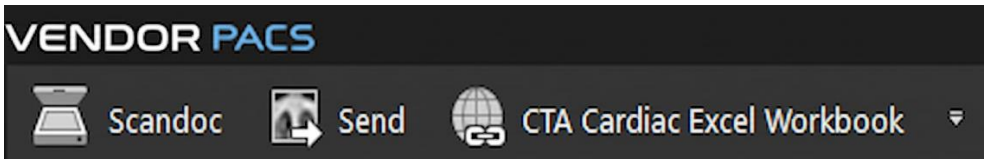


Tip: Before saving the workbook, highlight cell B9 so that when the user opens the workbook, their cursor is automatically placed in the Height cell to begin data entry.

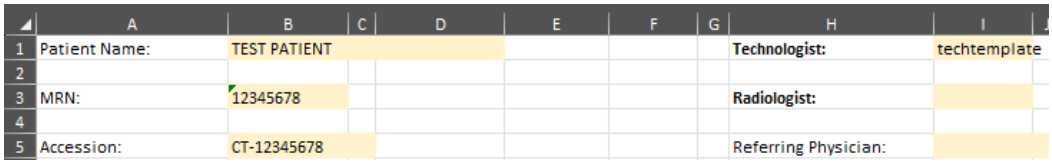
Workflow:

1. Select (highlight) CT exam on PACS worklist

2. Launch 'CTA Cardiac Excel Workbook' from external links in PACS worklist toolbar



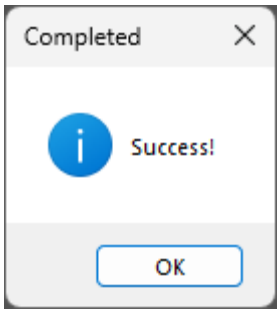
3. Workbook opens with Patient Name, MRN, Accession, Exam Date, and Technologist automatically inserted



	A	B	C	D	E	F	G	H	I
1	Patient Name:	TEST PATIENT						Technologist:	techtemplate
2									
3	MRN:	12345678						Radiologist:	
4									
5	Accession:	CT-12345678						Referring Physician:	

4. Fill out appropriate Input fields

   Two other field types - Calculated and Reference - are to be left as-is

5. Once finished, click Validate & Send button

6. Upon success, message is displayed and calculated values are available in SR mapping server for distribution to reporting system for auto population